POWER PULSE

Load Frequency Control Important Formulae

Notes by Balaji Sir

1 Governor and regulation

1. $$R = \frac{\Delta f / f_0}{\Delta P / P_{rated}} \ (\text{pu}), \qquad R\ (\text{Hz/MW}) = \frac{\%R \times f_0}{100\, P_{rated}}$$

2. $$\Delta P_g = \Delta P_{ref} - \frac{1}{R}\Delta f$$

speed-changer setting ΔP_ref

3. $$D = \frac{\Delta P_L}{\Delta f}\ (\text{MW/Hz}), \qquad B = \frac{1}{R} + D$$

area frequency response characteristic

2 Steady state

4. $$\Delta f_{ss} = \frac{-\Delta P_D}{1/R + D} \qquad (\text{single area})$$

5. $$\Delta f_{ss} = \frac{-\Delta P_D}{\frac{1}{R_1} + \frac{1}{R_2} + D_1 + D_2} \qquad (\text{two areas})$$

tie-line flow found from either area

6. $$\Delta P_i = -\frac{\Delta f}{R_i}$$

units on droop share a load change in proportion to rating/R

3 Dynamics and AGC

7. $$G_p(s) = \frac{K_p}{1 + sT_p}, \quad K_p = \frac{1}{D}, \quad T_p = \frac{2H}{fD}$$

power-system block

8. $$ACE = \Delta P_{tie} + B\,\Delta f$$

integral control drives ACE to zero

9. $$\text{With integral control: } \Delta f_{ss} = 0$$

primary control leaves a frequency error; secondary control removes it

10. $$\frac{dP_{tie}}{d\delta} = \frac{|V_1||V_2|}{X}\cos\delta_0 = P_s$$

synchronising coefficient of the tie-line